6 Probability

What students need to learn:

Ref	Content descriptor	Concepts and skills
SP m	Understand and use the vocabulary of probability and probability scale	• Distinguish between events which are; impossible, unlikely, even chance, likely, and certain to occur • Mark events and/or probabilities on a probability scale of 0 to 1 • Write probabilities in words or fractions, percentages or decimals
SP n	Understand and use estimates or measures of probability from theoretical models (including equally likely outcomes), or from relative frequency	• Understand and use estimates or measures of probability, including relative frequency • Use theoretical models to include outcomes using dice, spinners, coins • Find the probability of successive events, such as several throws of a single dice • Estimate the number of times an event will occur, given the probability and the number of trials
SP o	List all outcomes for single events, and for two successive events, in a systematic way and derive relative probabilities	• List all outcomes for single events, and for two successive events, systematically • Use and draw sample space diagrams
SP p	Identify different mutually exclusive outcomes and know that the sum of the probabilities of all these outcomes is 1	• Add simple probabilities • Identify different mutually exclusive outcomes and know that the sum of the probabilities of all these outcomes is 1 • Use $1 - p$ as the probability of an event not occurring where p is the probability of the event occurring • Find a missing probability from a list or table
SP q	**Know when to add or multiply two probabilities: when A and B are mutually exclusive, then the probability of A or B occurring is $P(A) + P(B)$, whereas when A and B are independent events, the probability of A and B occurring is $P(A) \times P(B)$**	• **Understand conditional probabilities** • **Understand selection with or without replacement**

Higher

Ref	Content descriptor	Concepts and skills
SP r	**Use tree diagrams to represent outcomes of compound events, recognising when events are independent**	• **Draw a probability tree diagram based on given information (no more than 3 branches per event)** • **Use a tree diagram to calculate conditional probability**
SP s	Compare experimental data and theoretical probabilities	• Compare experimental data and theoretical probabilities
SP t	Understand that if they repeat an experiment, they may – and usually will – get different outcomes, and that increasing sample size generally leads to better estimates of probability and population characteristics	• Compare relative frequencies from samples of different sizes